

Elucidation of thermo-mechanical properties of silicon nanowires from a molecular dynamics perspective

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ABSTRACT

The thermo-mechanical properties of silicon nanowires are of great importance in the state of the art silicon technologies. Herein, thermal and mechanical properties of silicon nanowires along [110] direction are illustrated by a molecular dynamics approach using Tersoff potential. Different lengths ranging from 10 to 45 nm and cross section widths from 2.2 nm to 6.5 nm as well as different temperatures from 200 to 500 K were studied. In all the cases, the corresponding thermal conductivities of silicon nanowires were found to be much smaller than the bulk silicon counterpart, in the range of $\sim 1\text{--}5\text{ W/(m.K)}$, and strongly dependent on size while less sensitive to temperature. The elastic modulus, fractural strain and stress were also investigated upon various size and temperatures indicating dramatically lower properties at higher temperature and as a result of nano-scaling. The presented findings on <110> silicon nanowires sheds further lights on the current understandings on thermo-mechanical behavior of silicon nanowires for applications such as in Li-ion batteries and provide complementary information for the relevant data bank.

1. Introduction

The rise of carbon nanotubes as emerging 1D nanomaterials in the late of 20th century [1–4] ignited many research studies throughout the world as a result of their exotic thermal, mechanical as well as electrical properties [5–11]. As such, 1D nanostructures have been under examination to investigate their thermal and mechanical properties for the development of future nano-based technologies [12–17]. Particularly, silicon nanowires (Si NWs) are key nanostructures that have received a great deal of attention as they are compatible with the conventional Si-based electronic technology [18,19] and offer potentially broad applications in many devices such as nanoelectromechanical systems (NEMS) [20,21], transistors [22], thermoelectric devices [23], sensors [24] and lithium-ion battery anodes [25,26].

By reducing the size of electronic components and increasing their density in electronic equipment, a lot of heat loads are imposed on the system, which necessitates a unique thermal management to minimize this heat load [27]. In addition, as the size approaches nanoscale, thermal behaviors such as hotspots are observed, the precise understanding of which improves the mechanical and electrical design of the devices [28]. The thermal properties of Si NWs are indeed affected by

their size due to the large surface-to-volume ratio as compared to the bulk silicon, rendering the corresponding thermal conductivities smaller than the bulk as the size effect limits the phonon transfer [29,30]. Therefore, understanding and predicting the thermal properties of Si NWs is of vital importance for equipment design. Li et al. [31] showed experimentally that the thermal conductivity of Si NWs with diameters of 22, 37, 56 and 115 nm is temperature dependent and two orders of magnitude lower than the bulk value. Nonetheless, experimental measurement of thermal properties has limitations and the high costs of the measuring instruments have urged the implementation of theoretical approaches.

Molecular dynamics (MD) simulation is a numerical method for predicting the thermal and mechanical properties of nanostructures. In general, two methods are available to obtain the thermal properties based on MD simulations. The equilibrium molecular dynamics (EMD), also known as the Green-Kubo method, calculates the thermal conductivity using the Fluctuation Dissipation and Linear Responding Theory. In contrast, the non-equilibrium molecular dynamics (NEMD) gives the thermal conductivity of nanowires using the well-known Fourier law [32]. For example, Volz and Chen [33] computed the thermal conductivity of Si NWs with cross-sections ranging from 2.58 to 28.62 nm² over

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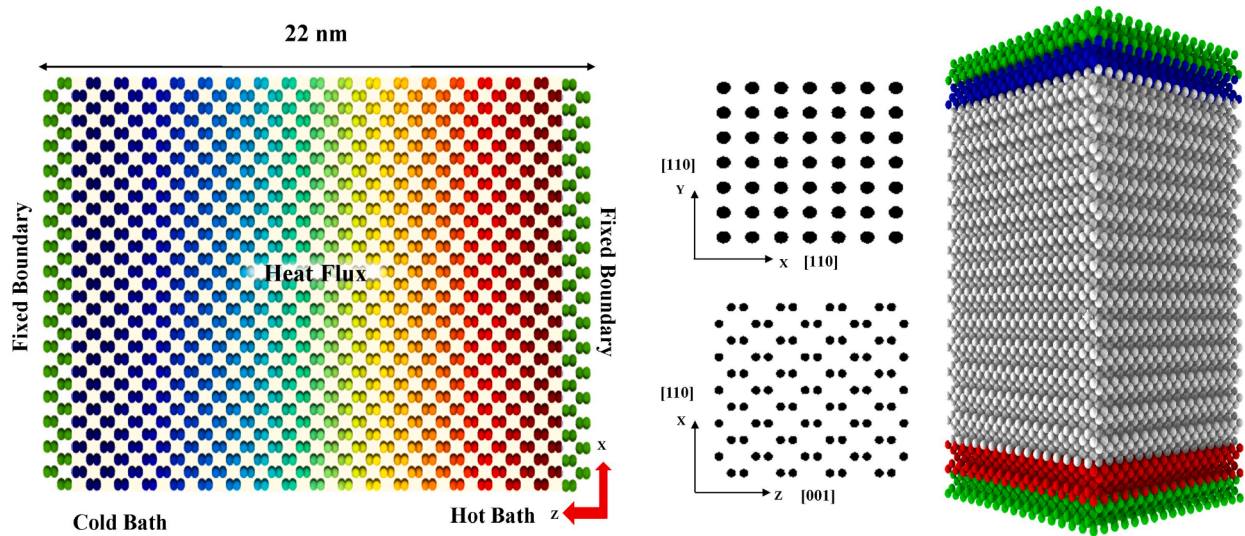


Fig. 1. System setup for NEMD thermal conductivity calculations of silicon nanowires. A silicon nanowire with $L = 22$ nm is shown as example.

the temperature range of 200–500 K by EMD simulation. Their calculations revealed a one to two orders of magnitude reduction in the thermal conductivity of nanowires as compared to their bulk counterparts. Soleimani et al. [34] employed both EMD and NEMD approaches to investigate the effect of length, width and temperature on the thermal conductivity of Si NWs. Their results showed that the process of thermal conductivity is similar in both methods albeit their values are different.

Apart from thermal properties, the mechanical properties of Si NWs are also needed to be taken into account in applications such as in Li-ion battery anodes where they undergo severe volume change over charge-discharge cycles [35] and in NEMS devices as in high-frequency electromechanical oscillators [36]. Nasr Esfahani [37] showed the effect of surface stress on the tensile behavior of Si NWs along the $\langle 100 \rangle$ direction with various geometric parameters at 10 K addressing the controversy between theoretical and experimental studies. Using NEMD simulations, Zhou et al. [38] reported the mechanical properties and deformation mechanism of Si NWs under pure bending, focusing on the effect of nanowire diameter, orientation and cross-sectional shape. Their results revealed that the nanowire diameter and cross-sectional shape have no effect on the final mechanism but the orientation does. Kang and Cai used the modified embedded-atom-method (MEAM) potential to investigate the failure mechanism of $[110]$ oriented-Si NWs under tensile tension at different temperatures up to 1000 K and demonstrated the size dependence of the fracture mechanism in Si NWs where their results well matched with experimental observations [39].

In spite of these efforts, a comprehensive evaluation on the thermomechanical properties of Si NWs with regards to their geometrical parameters, and temperature is yet needed to be elucidated. More specifically, $[110]$ oriented Si NWs have not been yet systematically explored in terms of both mechanical and thermal behaviors using Tersoff potential which is well-known for its good description of silicon material [40,41], to best of our knowledge. Herein, the thermal conductivity of Si NWs are first evaluated by NEMD simulation upon Tersoff potential. The nanowires are oriented along the $[110]$ direction which gives good stability against destruction. The Si NWs are constructed and studied at different lengths, widths and temperatures, and the corresponding results are corrected to take into account the quantum effects. The phonon density of states is also used to determine the intensity of phonons and understand the thermal transport features. In the following, mechanical properties of the nanowires including Young's modulus, fracture strain and fracture stress together with the effect of temperature, width and length on these properties are reported. The results confirm a dramatic reduction in both thermal and mechanical properties upon nanoscaling for Si NWs.

2. Theory and methods

2.1. Thermal properties

All the Si NWs were defined to be in the $[110]$ direction along the length, as shown in Fig. 1. To calculate the thermal conductivity, the green areas of atoms at the top and bottom were frozen to prevent the nanowires from rotating during the simulation. The red and blue areas are hot and cold baths, respectively, that apply heat flux to the system. A $2.2 \times 2.2 \times 22$ nm nanowire with a lattice constant of $5.431 \text{ \AA}$ in Z direction, contains 10,752 silicon atoms (Fig. 1).

The Tersoff potential [42] was selected to describe the interaction between silicon atoms. Considering the free boundary conditions in the X and Y direction and the periodic boundary condition along the Z direction, the nanowires were first relaxed under NPT ensemble for 2,000,000 steps ($\Delta t_{\text{step}} = 1$ fs) and then allowed to reach equilibrium at 300 K under the canonical ensemble by a Nose-Hoover thermostat for 2 ns followed by another 3.1 ns for data production under NEMD conditions. All the simulations in this study were performed by Large Scale Atomic/Molecular Massively Parallel Simulator (LAMMPS) package [43].

In the NEMD approach, the thermal conductivity (k) is calculated using the Fourier's law [44]:

$$k = -J/(dT/dx) \quad (1)$$

where J (W/m^2) is the heat flux and dT/dx is the temperature gradient between the hot and cold baths, the temperature of which are set to $T + 20$ and $T - 20$, respectively, with T equals to 300 K. After reaching the steady state conditions, the temperature gradient along the length of Si NWs can be calculated due to the heat flux.

2.1.1. Quantum correction

MD simulation examines the interatomic forces between atoms and molecules in terms of the principles of classical mechanics. The results of MD simulations at high temperatures (in the case of solids above the Debye temperature) are highly accurate as all vibrational modes are excited [45]. However, when the temperature decreases (for solids below the Debye temperature) quantum effects cannot be ignored. To compensate for this, quantum corrections are used. In fact, quantum correction depicts the values obtained from the classical system as compared to the corresponding values in the quantum system [46].

A simple approximation for calculating quantum correction is to rescale the mean temperature of the MD (T_{MD}) to the temperature that

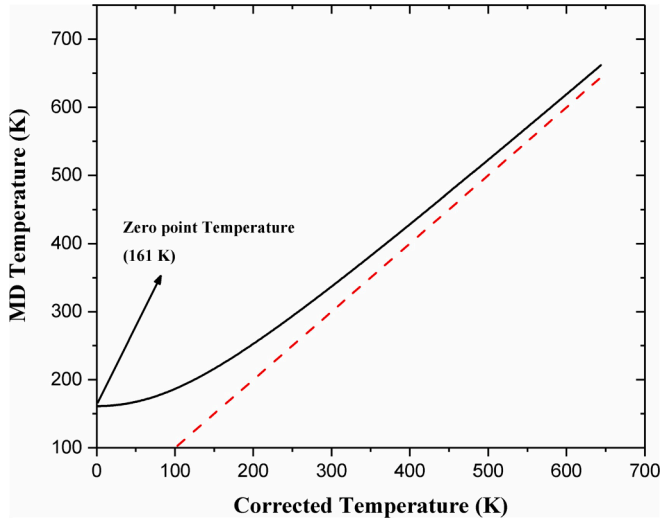


Fig. 2. MD simulation temperature versus quantum correction temperature for Si NWs. The dashed line shows the iseline for comparison.

includes the quantum effects by equating the total energy in the classical and quantum systems according to [34]:

$$3Nk_B T_{MD} = \int_0^{\omega_D} D(\omega) \left(\frac{1}{2} + n(\omega, T) \right) \hbar \omega d\omega \quad (2)$$

where, N is the number of atoms, k_B is the Boltzmann constant, T_{MD} is the simulation temperature, $D(\omega)$ is the phonon density of states, ω is the phonon frequency, ω_D is the Debye frequency cut off, and $n(\omega, T)$ is the phonon occupancy number, which is given by the Planck distribution function. Considering a one dimensional density of states for the phonons, equation 2 is simplified to:

$$T_{MD} = \frac{T_D^2}{T_D} \int_0^{\frac{T_D}{T}} \frac{x}{e^x - 1} dx + \frac{T_D}{4} \quad (3)$$

where T_D is the Debye temperature. Fig. 2 shows the relationship between the MD and quantum corrected temperatures for Si NWs. When the system temperature is low enough, the temperature of the quantum correction is much higher than the statistical temperature obtained from the MD simulation and it must be corrected. It also shows that quantum effects are small when the temperature is above 400 K.

As shown in Fig. 2, the minimum temperature that could be

classically studied is actually 161 K. In fact, this temperature corresponds to 0 K based on the quantum corrections and any calculations beyond that limit is not valid. For this reason, all the simulation were performed above 200 K in this study.

The corrected thermal conductivity can be then obtained from equation 4 for structures whose thermal conductivity are calculated at a temperature lower than the Debye temperature:

$$k_{qc} = k_{MD} \frac{\partial T_{MD}}{\partial T} \quad (4)$$

According to Eq. (4), any thermal conductivity obtained from MD calculations based on classical theories at a given temperature could be assigned to its quantum-corrected counterpart at the modified temperature rather than T_{MD} which is also considered in this study.

2.2. Mechanical properties

To obtain the mechanical properties of Si NWs using LAMMPS package, Tersoff potential at 300 K and the time step of 1 fs were applied. The boundary conditions were the same as in the case of thermal properties. Before applying a tensile load to the nanowire, the sample structure is first balanced by a Nose-Hoover barostat and thermostat for 2.5 ps. Then the strain as $\Delta\epsilon = \Delta L/L_0$, where L_0 is the initial length of the nanowire and ΔL is the change in length due to the external strain, is applied to the system. The nanowire's length increases due to the applied strain at a constant strain rate of $1 \times 10^{-9} s^{-1}$.

The obtained stress-strain curves include an initial linear region and then a nonlinear behavior up to the ultimate tensile strength (UTS). Further increase in strain above the UTS point results in a sudden drop in stress values, leading to the rupture of the nanowire.

Stress components (S) were calculated using virial theory [47]:

$$S = \frac{1}{V} \sum_{a \in V} \left[-m \vec{v}_a \otimes \vec{v}_a + \frac{1}{2} \sum_{a \neq b} \left(\vec{r}_{ab} \otimes \vec{F}_{ab} \right) \right] \quad (5)$$

where m and $\vec{v}_a$ represent mass and velocity vector of the atom a , $\vec{r}_{ab}$ is the position vector of the atom a with respect to atom b , $\vec{F}_{ab}$ is the force vector on atom a caused by atom b , V the volume of the silicon nanowire and $\otimes$ denotes the Tensor product.

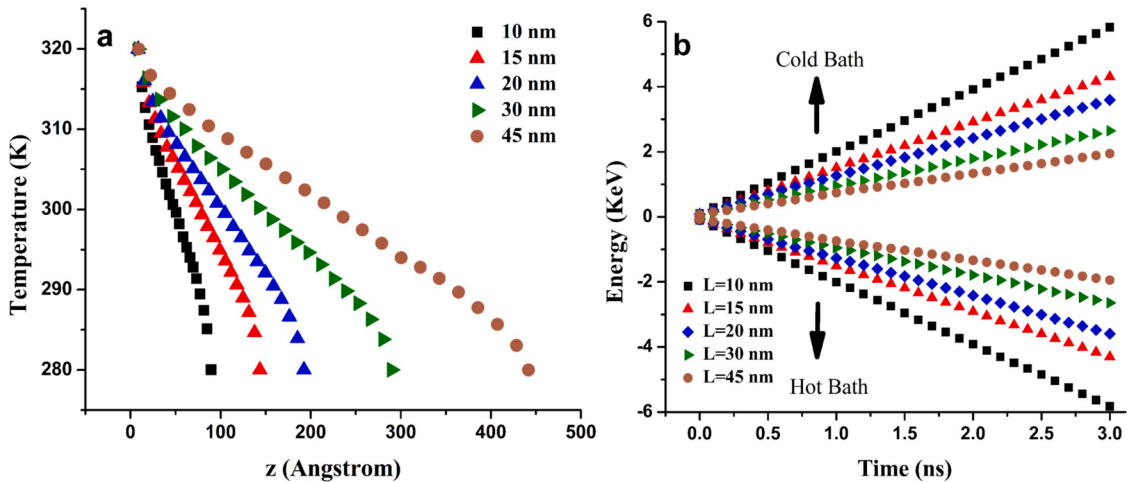


Fig. 3. a) Temperature profile of the silicon nanowires. b) Accumulate kinetic energy added into and removed from the hot and cold baths, respectively. Cross-section width = 4.3 nm, $T = 300$ K.

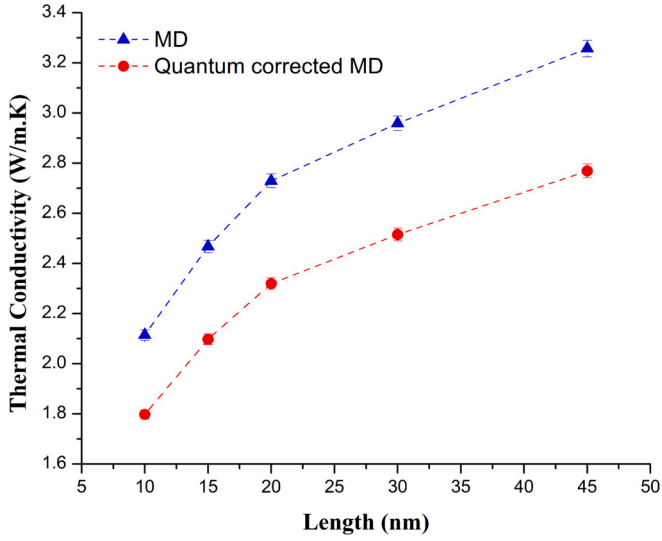


Fig. 4. Thermal conductivity of the silicon nanowire for a cross-section width of 4.3 nm as a function of length at 300 K.

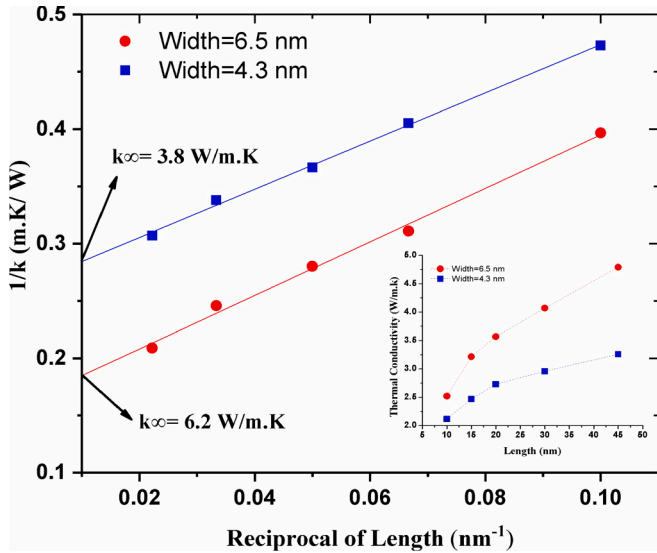


Fig. 5. Reciprocal of thermal conductivity versus reciprocal of length for silicon nanowires.

3. Results and discussion

3.1. Thermal conductivity of Si NWs

3.1.1. Effect of length

To calculate the thermal conductivity, the system must be in a steady state to eliminate the initial stresses caused by the unbalanced process between hot and cold baths.

Fig. 3a shows the temperature profiles for a silicon nanowire with a fixed width of 4.3 nm but different lengths of 10, 15, 20, 30 and 45 nm. It could be observed that the temperature distribution in the middle of the nanowire is linear in all the cases confirming the steady-state conditions. Fig. 3b shows the cumulative energy that is reduced from the hot bath and added to the cold bath during the NEMD simulation. The slope of each line gives the corresponding heat flow generated/received by the hot/cold baths. By dividing this energy to the cross-sectional area, the heat flux would be deduced.

The results of MD calculations for different lengths are shown in

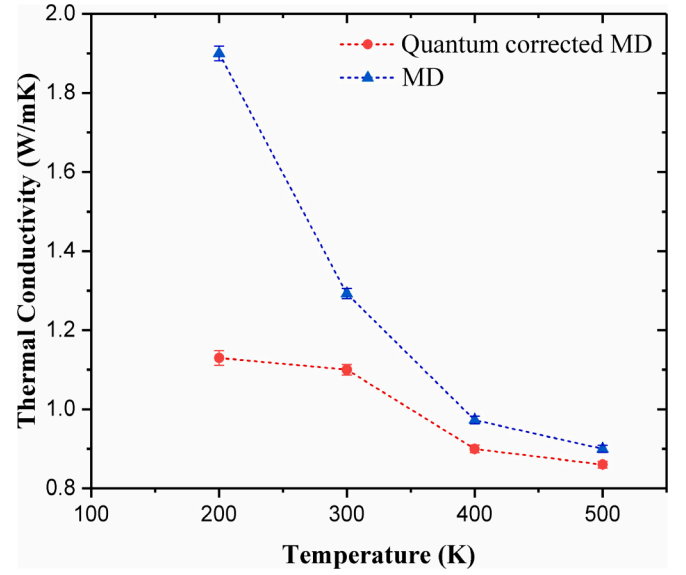


Fig. 6. Thermal conductivity of the silicon nanowire for 2.2 nm cross-section width with the length of 20 nm versus Temperature.

Table 1

Thermal conductivities based on MD and quantum corrected MD for a silicon nanowire with a cross-section width of 2.2 nm.

T _{MD} (K)	T (K)	k _{MD} (W/mK)	k (W/mK)	k _{MD} /k
200	125	1.90	1.13	1.68
300	258	1.29	1.10	1.17
400	370	0.97	0.90	1.08
500	477	0.90	0.86	1.05

Fig. 4. It is clear that the thermal conductivities are much lower than bulk silicon and by increasing the length, the thermal conductivity also increases.

When the nanowire length is much shorter than the bulk, they are comparable to the phonon mean free path (MFP) which carries heat in silicon. As a result, significant scattering of phonons occurs, which reduces the thermal conductivity. In this study, a maximum length of 45 nm has been considered for the nanowires which is yet much smaller than the phonon mean free path of bulk silicon (300 nm) at 300 K [48]. Therefore, as the length of the nanowire increases in the range of 10–45 nm, the corresponding thermal conductivity dramatically increases due to the enhancement in the effective phonon MFP which is even more intense for shorter nanowires.

As depicted in the inset of Fig. 5, we further conducted the simulations for Si NWs with a cross-section width of 6.5 nm which presented a similar trend confirming the above discussions. Fig. 5 shows that the reciprocal of MD thermal conductivity is linearly related to length for both widths of 4.3 and 6.5 nm according to the following equation [49]:

$$\frac{1}{k} = \frac{a_0^3}{4k_B v} \left(\frac{1}{l_\infty} + \frac{4}{L_z} \right) = \frac{1}{k_\infty} \left(1 + \frac{4l_\infty}{L_z} \right) \quad (4)$$

where, a_0 is the lattice parameter, v is the group velocity of phonons, l_∞ is the phonon MFP of a nanowire with infinite length, L_z is the nanowire length and k_B is the Boltzmann constant. By extrapolating the fitting line to zero, one can obtain the thermal conductivity for a silicon nanowire with infinite length (k_∞).

The corresponding values for width of 4.3 nm and 6.5 nm are 3.8 and 6.5 W/m.K, respectively. Using equation 4, a phonon MFP of 2.0 and 3.6 nm would be deduced for these nanowires, respectively. As the surface to volume ratio is lower for the latter, less phonons are diffusely scattered from the surface and a longer phonon MFP is obtained for the cross

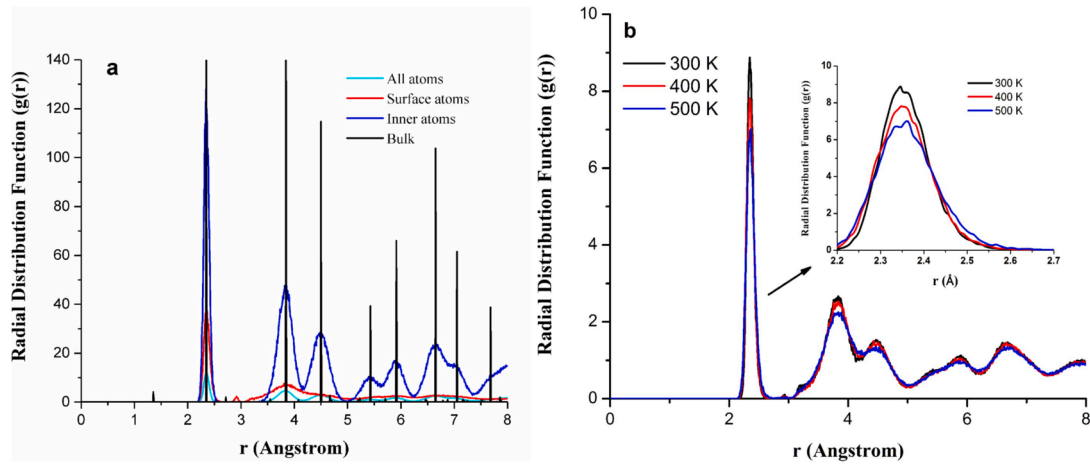


Fig. 7. a) Radial distribution function (RDF) of Si-Si for silicon nanowire (surface, inner and all atoms) and bulk silicon at 300 K. b) RDF of Si-Si for silicon nanowire at 300, 400 and 500 K. The inset shows the enlarged area. The length and cross section width are 22 and 6.5 nm, respectively.

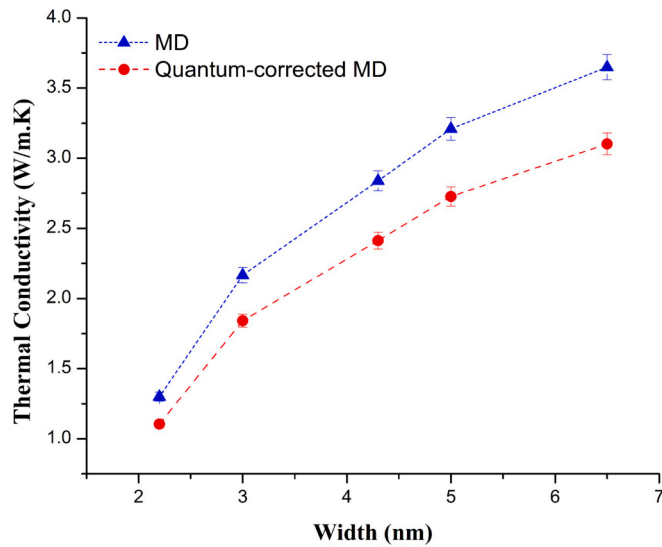


Fig. 8. Thermal conductivity as a function of cross-section width for si nano-wire. $T = 300$ K; $L = 22$ nm.

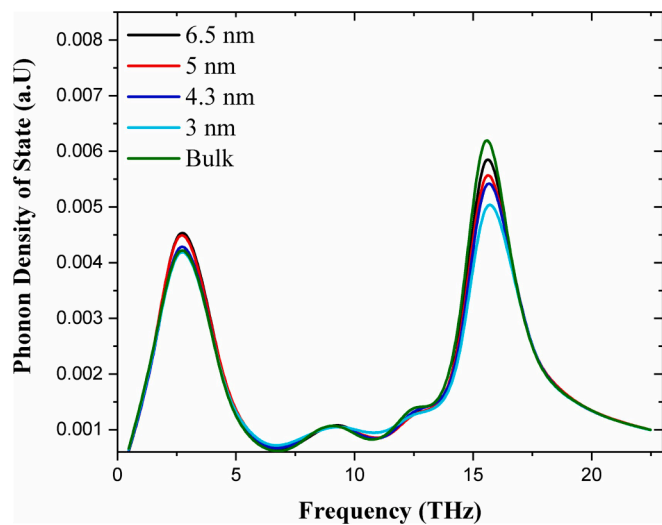


Fig. 9. Phonon density of states (DOS) for silicon bulk and silicon nanowires with different width. $L = 22$ nm, $T = 300$ K.

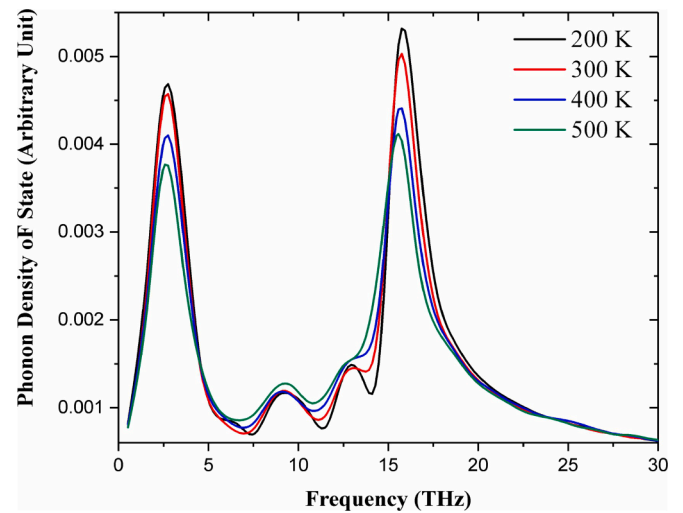


Fig. 10. Phonon density of states (DOS) of the silicon nanowire at different temperatures. $L = 22$ nm; width = 2.2 nm.

section width of 6.5 nm which gives rise to higher thermal conductivities as compared to the 4.3 nm width counterpart. Our findings are in good agreement with the results of Wang et al. [48] and Soleimani et al. [34] using Stillinger-Weber potential for Si NWs.

3.1.2. Effect of temperature on the thermal conductivity of Si NWs

The dependency between thermal conductivity and temperature in Si NWs is shown in Fig. 6 in that the thermal conductivity decreases with increasing temperature. Due to the scattering of phonon-phonon and Umklapp, which is common at high temperatures, the phonon MFP decreases [41]. In fact, the population of scattered phonons increases by increasing the temperature resulting in decrement of thermal conductivity. As expected, for temperatures close to the Debye temperature (i.e. 400–500 K), the corresponding thermal conductivities for quantum-corrected calculations are consistent with that of classical MD for infinitely long nanowires. Nonetheless, at low temperatures (i.e. <400 K), the difference between thermal conductivities in classical and quantum modes becomes significant as the density of states for phonons presented in Eq. (5) follow the Debye model for quantum corrections while the classical MD calculations are based on the equipartition law (Table 1) [34].

At high temperatures, it is also possible to observe disorder in the crystal structure. For this purpose, a radial distribution function (RDF)

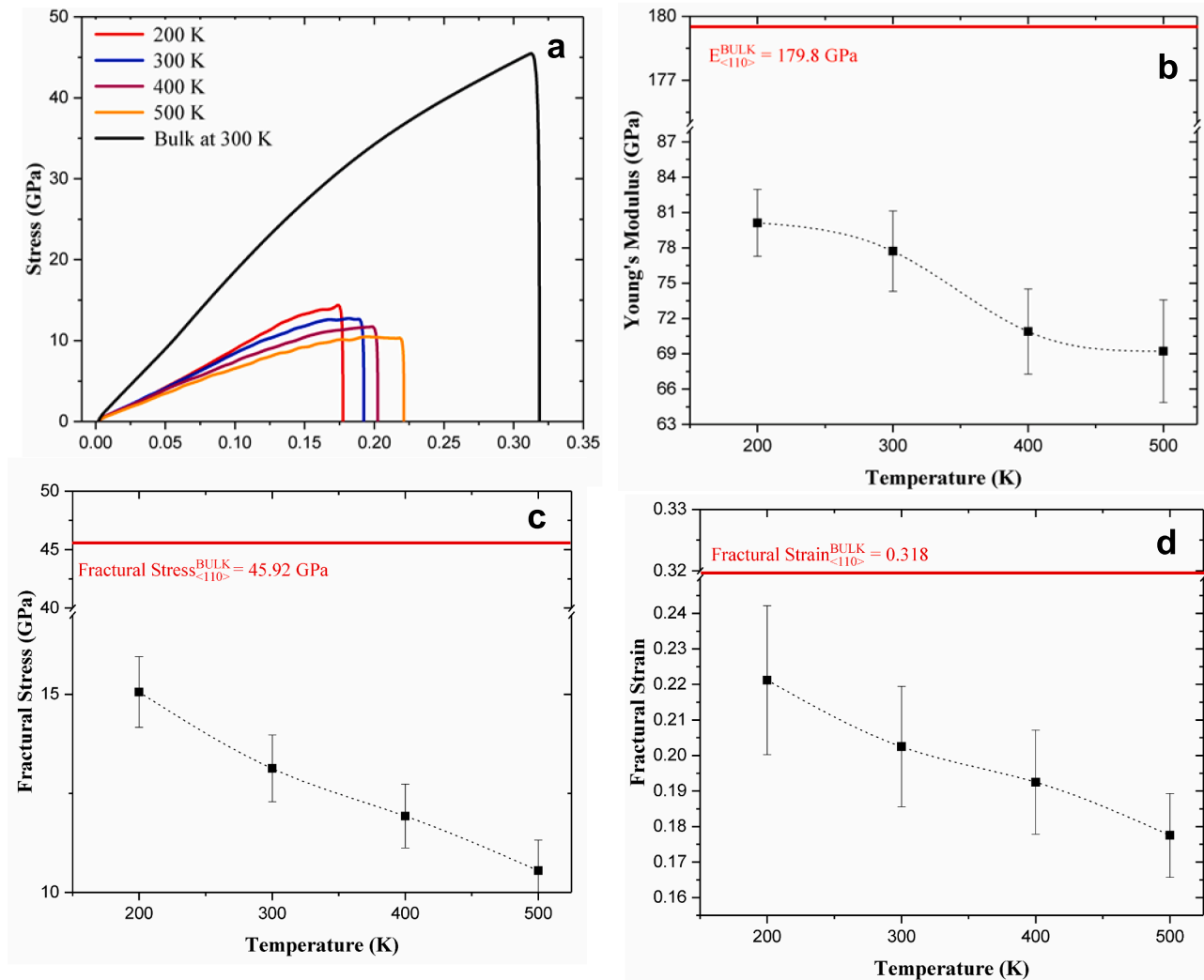


Fig. 11. (a) Stress–strain curves; (b) Young modulus; (c, d) fractural stress and strain of the silicon nanowire at different temperatures. $L = 22$ nm; width = 2.2 nm.

was employed to show the nature of silicon crystals in silicon nanowires. Fig. 7a shows the RDF for three different groups of atoms in the nanowire (surface atoms, inner atoms, and all atoms) together with the silicon bulk at 300 K. For silicon bulk, there is a sharp peak at 2.35 Å. This peak is associated with the distance of the first neighborhood of silicon atoms and therefore the complete crystallinity of silicon. In the case of the nanowire, the behavior of the internal atoms is much like to that of silicon bulk but as it approaches surface atoms, the intensity of the peaks dramatically drops while their position does not change confirming the surface state with missing atoms. The decreased peak intensities for the all atoms in the nanowires as compared to bulk silicon is due to the significantly increased surface to volume ratio.

On the other hand, as the temperature increases, the peaks intensity in the RDF diagram decreases compared with that of at 300 K, as shown in Fig. 7b. This observation is attributed to the movement of atoms at high temperatures and their departure from the equilibrium position in the lattice which enhances the degree of disorder in the silicon nanowire lattice.

3.1.3. Effect of width on thermal conductivity

To investigate the effect of different widths on thermal conductivity, nanowires were simulated at different cross-section widths of 2.2, 3, 4.3, 5, 6.5 nm but a fixed length of 22 nm.

As shown in Fig. 8, the thermal conductivity of Si NWs increases upon increasing the width owing to the smaller surface-to-volume ratio.

By reducing this ratio, the boundary effects and consequently scattering of the phonons at the boundary are reduced and phonons can carry energy over longer distances.

3.1.4. Phonon density of states

In solid state physics, the density of states (DOS) of a system describes the amount of states that can be occupied by the system at any energy level. If the DOS at a particular energy has the highest value, it means that there is a lot of space to be occupied by the system. The heat transfer properties of nanowires depend on the vibrational phonon DOS. The DOS for silicon nanowire-all atoms is obtained from the following equation [34]:

$$D(\omega) = \int e^{i\omega t} \frac{\langle V(0)V(t) \rangle}{\langle V(0)V(0) \rangle} dt \quad (5)$$

Using the Fourier transform of the Velocity Autocorrelation Function (VACF), the phonon DOS of Si NWs is calculated. Fig. 9 shows the corresponding phonon DOS for Si NWs with different widths and bulk silicon. It could be observed that when the width size decreases, the transverse optical (TO) mode height, which is in the range of 14–18 THz, also decreases and it slowly shifts to higher frequencies. In addition, in the frequency range of 2–5.5 THz, the transverse acoustic (TA) mode shifts to lower frequencies, while the height of TA modes does not change much as compared to TO. The sharp changes in these two regions can be attributed to the increase in the number of surface atoms due to

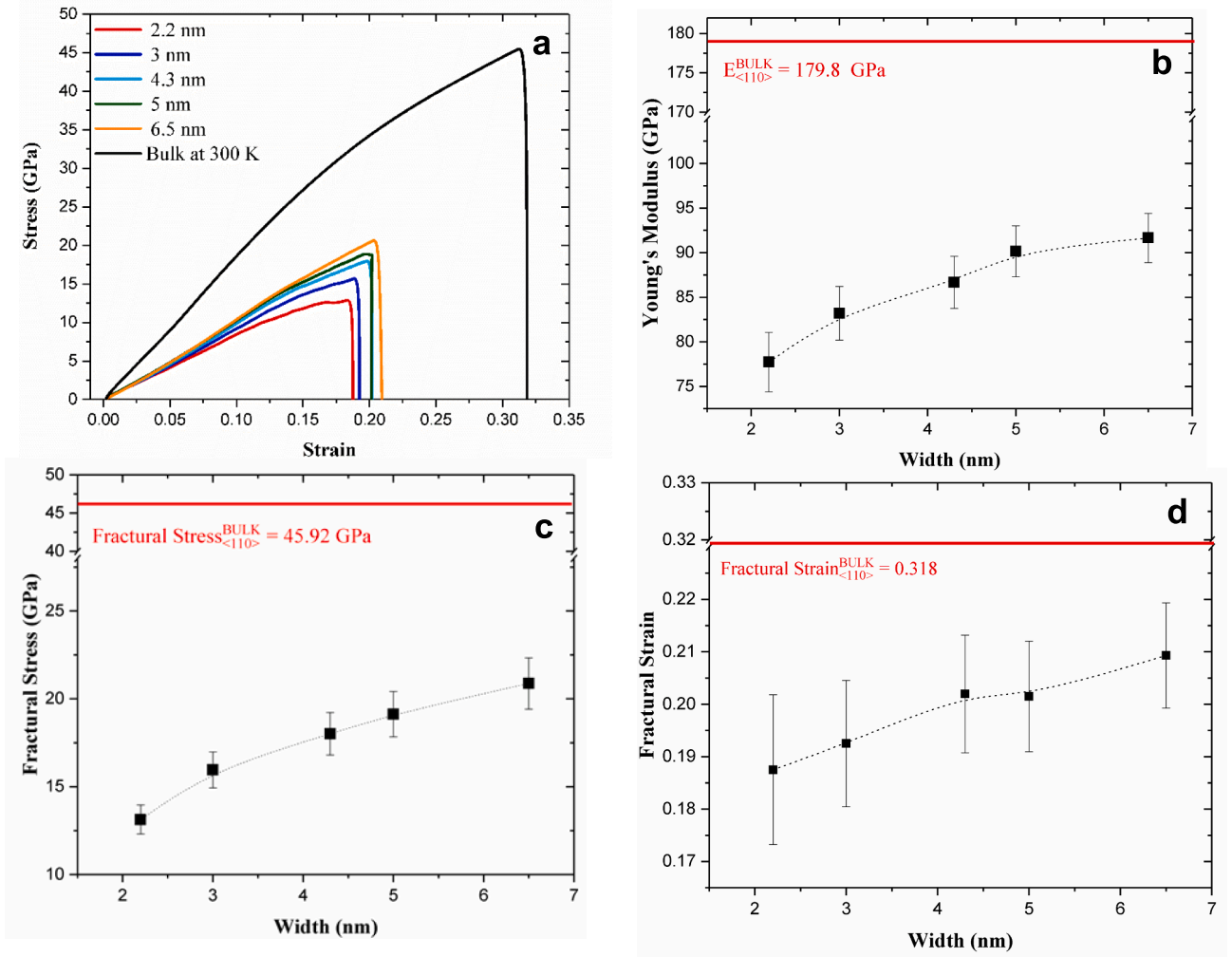


Fig. 12. Stress-strain curves; Young modulus; fracture stress and strain of silicon nanowire with 22 length and different width at 300 K.

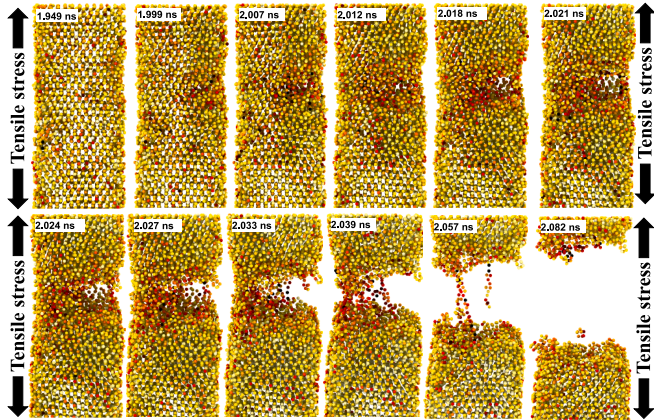


Fig. 13. Snapshots from animation of a typical failure of a silicon nanowire under tensile load. L = 22 nm; width = 3.0 nm, T = 300 K. The light to dark colored spheres show the silicon atoms under low to high tension.

the enhancement in the surface-to-volume ratio or the decrease in the bonding force on the surface atoms, which causes the TA mode to move towards low-frequency region.

The phonon DOS were also calculated for a nanowire at different applied temperatures. Fig. 10 illustrates that upon increasing the temperature, the intensity of the peaks drops and both the TA and TO modes

(low and high frequency zones) shift towards lower frequencies. Irregularities and displacements that exist at high temperatures increase the disturbance and scattering of the phonon and as a result, the tendency to lower frequencies increases. Further, the DOS shows a decrease from 0.00464 to 0.00377 at ~2.5 THz for TA mode and from 0.00513 to 0.00412 at ~15.5 THz for TO mode as the temperature rises from 200 K to 500 K indicating a smaller number of vibrational modes for phonons at higher temperatures and as a result a lower thermal conductivity, which is also consistent with the computed results.

3.2. Mechanical properties

3.2.1. Dependence on temperature

As shown in Fig. 11a, b, the yield strength and elastic modulus of Si NWs decrease with increasing the temperature. The total kinetic energy of all the atoms in the system is obtained from the following equation [50]:

$$E_K = \sum_{i=1}^N \frac{1}{2} m v_i^2 = \frac{3}{2} N k_B T \quad (6)$$

where E_K is the total kinetic energy; N is the total number of atoms; k_B is the Boltzmann constant and T is the thermodynamic temperature.

According to Eq. (6), as the temperature goes up, the total kinetic energy in the system proportionally increases resulting in a faster movement of atoms. Under an external load, the attractive force between the atoms decreases and the atoms are departed from their

equilibrium state. Therefore, the corresponding stress at a fixed strain drops for the nanowires and the elastic modulus decreases accordingly.

At high temperatures, lattice defects in the nanowire are created as the atoms displace from their equilibrium positions. As a result, the number of defects and crystal disorders of the nanowire increases upon increasing the temperature, which in turn gives rise to a lower yield strength at high temperatures. This is also valid in the case of fractural stress and strain as shown in Fig. 11c, d, respectively.

3.2.2. Dependence on width

Si NWs with different cross section widths were also simulated to investigate the effect of size on the mechanical properties. Fig. 12a-d respectively shows the stress-strain curves, the modulus of elasticity and the corresponding fractural stress and strain values at different widths. It is clearly comprehended that the mechanical properties of Si NWs strongly depend on the size.

Upon increment of the cross section width in the range of 2.2–6.5 nm, the corresponding elastic modulus is enhanced (Fig. 12a, b). A same trend also applies to the stress and strain of the breaking point as depicted in Fig. 12c, d, respectively. Atomic effects are clearly evident for nanowires with cross-sectional widths less than 10 nm. The modulus of elasticity for the nanowires are much lower than the silicon bulk counterpart with 179.8 GPa. In fact, in the case of the largest width studied here, the corresponding modulus elasticity is only about half of the bulk value. As the cross-sectional area increases, the elastic modulus of the nanowire converges to that of the bulk material. Therefore, the mechanical strength of the silicon nanowires are improved as a result of decreasing the ratio of the surface to total atoms.

Fig. 13 shows snapshots from animation of a silicon nanowire with a width of 3.0 nm and a length of 22 nm under tensile load which clearly reveals the nucleation of crystal defects in the middle parts. At prolonged time steps (>2.01 ns), the defects are developed in the form of pores resulting in missing bonds between the silicon atoms in the nearby. Accordingly, the pores grow further which gives rise to the rupture of the material from one side followed by complete failure of the nanowire.

4. Conclusion

In summary, we systematically investigated the thermomechanical properties of [110] silicon nanowires using Tersoff potential. The effect of length, cross-section width, and temperature on the properties were elucidated. By reducing the size of the nanowire the thermal conductivity also decreased.

The corresponding phonon mean free paths for the widths of 4.3 nm and 6.5 nm were 2.0 nm and 3.6 nm, respectively, which correspond to the thermal conductivities of 3.8 and 6.2 W/m.K, respectively. The radial distribution function of the nanowires demonstrated that the presence of disorders is evident for the silicon nanowires due to the increased surface to volume ratio resulting in lower thermal conductivities as compared to the bulk counterpart. The reduction in thermal conductivities was also analyzed by the phonon density of states showing suppressed peaks for lower cross section widths in both transverse acoustic and optical branches. The classical MD showed a rise in the thermal conductivity from 0.9 to 1.9 W/m.K by decreasing the temperature from 500 K to 200 K which was also confirmed by the radial distribution function while the quantum-corrected calculations revealed that the thermal conductivity is less sensitive to temperature and approaches to 1.13 W/m.K.

The properties of silicon nanowires were also evaluated under tensile load. As the temperature of the system increased, the Young modulus dropped from 80.1 GPa to 69.2 GPa. Moreover, it was confirmed that nanoscaling dramatically reduces the corresponding mechanical properties of silicon nanowires as compared to the bulk material. The thermo-mechanical findings of this study provide more in-depth understandings for the future applications of Si nanowires in the state of

the art silicon based technologies.

5. Data availability statement

The raw/processed data required to reproduce these findings cannot be shared at this time as the data also forms part of an ongoing study.

CRediT authorship contribution statement

Farzane Hasheminia: Investigation, Data curation, Writing – original draft. **Yaser Bahari:** Conceptualization, Supervision, Writing - review & editing, Project administration. **Ali Rajabpour:** Methodology, Supervision, Writing - review & editing. **Saeed Arabha:** Investigation, Data curation, Writing – original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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